

Uijin Cho

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EDUCATION

Brown University, *B.S. Computer Science & Applied Mathematics (GPA: 4.0)*

Expected: May 2029

- **Activities & Societies:** Hack@Brown, Brown Badminton Team, Providence Medical Orchestra, Rewriting the Code

SKILLS

- **Programming Languages:** Python, C/C++, SQL, JavaScript/TypeScript, Kotlin, Go, Bash, HTML/CSS
- **Quantitative & ML Tools:** PyTorch, XGBoost, NumPy/pandas, OpenAI API, RAG, pgvector, HPC computing, modeling
- **Technical Tools:** Git & GitHub, Docker, Linux, AWS, Firebase, FastAPI, Next.js/React, Vercel, Figma

EXPERIENCE

Brown University Freilich Lab

Providence, RI

Machine Learning Research Assistant

Jun 2026 – Present

- Extended an existing ML research pipeline to support deep learning, applying quantitative modeling to hindcast pollutant levels
- Processed 10+ years of multiband satellite imagery through an end-to-end Python pipeline, distributed across a HPC cluster
- Engineered a modular PyTorch training system with visualization outputs, designed for reproducibility in multiple configurations

GoDaddy

Tempe, AZ

Software Engineer Intern (Intern Hackathon Winner)

Jun – July 2026

- Migrated API calls to AWS services across 10+ internal Go microservices, improving cloud resource compatibility
- Won GoDaddy's Intern Hackathon leading integration of the FastAPI backend with Next.js/React frontend, unifying 7+ modules including an LLM-powered business advisor into a working end-to-end demo; presented product vision and demo to judges

The Brown Daily Herald

Providence, RI

Data Analyst

Jan – May 2026

- Engineered scalable Python data pipelines to clean, model, and visualize 70K+ course records spanning 10 years
- Identified trends in enrollment and faculty popularity across 2,100+ data points, translating records to support editorial decisions
- Developed data visualizations for campus poll, synthesizing employment by year and service approval metrics into clear reporting

Robolabs

San Jose, CA

Engineer & Robotics Coach

Mar 2024 – Aug 2025

- Implemented a closed-loop PID controller in C++ with real-time gain tuning, improving motor precision and system stability

PROJECTS

perennial: Symptom Tracker for Chronic Diseases (YHack 2026 Winner) | Kotlin, Firebase, XGBoost

- Engineered an Android & WearOS app in Kotlin with Firebase Auth and Cloud Storage, supporting chronic condition logging
- Trained and personalized an XGBoost pain-level classifier with 95.8% accuracy with warm-start fine-tuning for user adaptation

The Lot: Drum Corps Audition Search RAG Tool | OpenAI API, pgvector, RAG, FastAPI

- Designed a PostgreSQL + pgvector schema for semantic search over embedded DCI documents, powering RAG retrieval
- Built a Python (FastAPI) ingestion pipeline to parse, chunk, and embed documents using OpenAI's text-embedding-3-small model
- Developed a RAG chat assistant: embedding-based retrieval, prompt construction, and profile-personalized generation

Image to Lithophane STL Generator | Python, OpenCV, React, Javascript

- Built a pipeline in Python, mapping pixel brightness to variable mesh thickness to generate 3D-printable STL files from an image
- Ported to a React web app for browser-based pipeline with no backend, supporting adjustable resolution, scale, and layer depth

LEADERSHIP

Hack@Brown, *Hackathon Co-Director (prev. Experience Organizer)*

Aug 2025 – Present

- Led a team of 40 students for Hack@Brown 2027, an event with 500 participants from around the world and \$50K+ in funding
- Coordinated cross-functional efforts across design, engineering, and sponsorship teams to maintain a cohesive brand identity
- Launched efforts to start the only collegiate-organized high school hackathon for 100 Rhode Island students

Blu's Hacks, *Hackathon Director 2024 & 2025*

Apr 2023 – Mar 2025

- Directed a beginner-friendly hackathon for Bay Area students, earning \$35K in sponsorships and increasing participation by 25%